IMPERIAL
SCHOOL OF PUBLIC HEALTH

Saudi Arabia

A Health Systems Analysis
24th March 2025

Health Systems Policy & Financing Assessment *– Emmanuel, Heather, Issy, Sena, & Carine*

Demographics & Epidemiology

Demographic Shifts

- 85% of Saudi Arabia’s population live in urban centres, resulting in advanced healthcare infrastructure in cities but disparities in rural areas. In addition, expatriates comprise 38% of the population.
- Saudi’s population is actually considered young, with 63% under 30 years of age. However, it’s experiencing a shift towards an aging population, with the proportion of people 60+ expected to rise from the current 4.4% of the population to 25%.

Demand & Investment

- Owing to to increasing life expectancies and declining fertility rates, the anticipated aging will lead to higher rates of chronic disease, increasing demand for specialized services and further necessitating a shift from the healthcare system’s current and historic focus on curative care to preventative care and chronic disease management.
- For this reason, Saudi is investing heavily in its healthcare infrastructure, (with spending projected to reach $77bln by 2030) focusing primarily on geriatric care and digital health solutions.

Political

Political transformation

- Shifting from a traditional absolute monarchy to a modern authoritarian system with still 100% centralised power within the royal family.
- The new regime is more open to innovation is directly translating into a more dynamic, efficient, and modern healthcare system.
- This centralised decision-making has enabled swift and large-scale healthcare reforms.
- However, issues within a centralised authoritarian system like accountability and corruption risks as well as transparency and censorship are still persistent.

Shift of Health System Focus - Vision 2030

- Saudi Arabia's health reform priorities include preventive care, digital transformation (such as the 937 Call Center and the Seha Virtual Hospital), Public-Private Partnerships (PPP), Saudization of workforce, mental health reform, medical tourism, reduced reliance on imported drugs, and enhanced decentralisation through greater autonomy for regional healthcare bodies.

Legal

Legal System

- Saudi Arabia's legal system blends Sharia law, royal decrees, and modern governance, with the king and royal family holding absolute authority over legislation, regulation, and judicial oversight under the 1992 Basic Law of Governance.

Reformation of Regulatory System

- Saudi Arabia’s regulatory environment is strict yet selective, heavily influenced by royal priorities.
- Vision 2030 combines centralised authoritarian control with modernisation efforts to attract investment and diversify the economy.
- Bureaucratic challenges and Nazaha's limited independence hinder regulatory progress.

Ministry of Health (MOH)	Primary regulator of public healthcare services, hospitals, and primary care centres.
Saudi Food and Drug Authority (SFDA)	Regulates pharmaceuticals, medical devices, and food safety.
Saudi Commission for Health Specialties (SCFHS)	Licenses healthcare professionals and accredits training programs.
Saudi Central Bank (SAMA)	Regulates health insurance financing and fintech initiatives.

Economic & Ecological

- Healthcare spending increased to ~14.4% of the budget (2023), focusing on infrastructure and digital health.
- Saudi Arabia's oil-driven economy (32% of GDP in 2023), though boasting a high GDP per capita, remains vulnerable to oil price fluctuations.
- While oil revenues have historically funded free public healthcare, the kingdom is now pursuing economic diversification, decreasing dependence on oil under Vision 2030.
- Extreme heat, poor air quality, water scarcity, and urbanisation contribute to Saudi Arabia's disease burden – resilience against climate change.

Economic Diversification Initiatives

- Prioritising non-oil sectors like tourism, tech, renewables, logistics, and mining.
- Public Investment Fund (PIF): Driving domestic projects (e.g., NEOM) and global investments.
- Fiscal & Energy Reforms: Introduced 15% VAT and excise taxes to boost non-oil revenue.
- Saudi Green Initiative: Targets 50% renewable energy by 2030 and net-zero emissions by 2060 (e.g., solar/wind farms).
- Public-private partnerships (PPPs) and privatization (e.g., Aramco’s IPO) to fund diversification.
- Expanding ports, railways, and air routes (e.g., King Abdullah Port) to enhance global trade connectivity.

Socio-Cultural

Regional Disparities in PHCs

- Saudi Arabia has notable regional disparities in the distribution of primary healthcare centres. According to a recent study analysing the distribution of PHCs across the country's 20 administrative regions, significant differences were observed in the number of PHCs available per 100,000 people.
- The capital, Riyadh, has a high ratio of 7.5 centres per 100,000 compared to many other regions, whereas Jeddah has only 3.2 despite being a populous city.
- Notably, an overall decline in the number of primary healthcare centres since 2017 suggests a growing strain on primary healthcare resources as the population grows, and will start to age.

Inequitable Access Contributes to Disparities in Health Outcomes

- This is concerning because uneven distribution is correlated with reduced accessibility to preventative care, chronic disease management, and early intervention in underserved regions, leading to downstream increases in mortality rates from preventable conditions.

Performance

	KSA	UAE	Oman	Qatar
Life expectancy at birth (years)				
Male	74.9	76.1	77.2	77.3
Female	77.3	78.3	79.0	79.4
Infant mortality (per 1,000 live births)	5.3	5.6	6.7	9.5
HALE at birth (years)	65.0	66.5	68.0	67.5
HALE at 60 (years)	15.2	16.0	16.5	16.0
Annual Total Health Expenditure (US$/Capita)	1,147	1,200	1,800	1,000
Health Expenditure as Percentage of GDP (%)	5.7	4.0	2.2	4.3
Physician-to-Population ratio (per 100,000 people)	3.077	2.5	3.7	2.0

Overview

- Saudi Arabia has a high life expectancy but compared to neighbouring countries is doing worse which may indicate a problem with healthcare system or living conditions compared other high income countries
- Whilst having a lower life expectancy compared to neighbouring countries, infant mortality is lower than neighbouring countries suggesting better prenatal and postnatal care.
- South of Arabia spends less per capita than Oman but more than Qatar which reflects a moderate investment in healthcare compared to neighbours but this does not represent efficiency and proper allocation of resources which plays a vital role
- Although looking at health expenditure as a percentage of GDP we can see that Saudi Arabia allocates a high percentage of its GDP to healthcare indicating a strong commitment to improving health outcomes.

- The country's digital health market's (projected to grow at a compound annual growth rate (CAGR) of 21.3%, reaching $13.3bln by 2031;) growth reflects the country's significant commitment to integrating technology into healthcare delivery, aligning with its 2030 goals (12).

- Integration of digital technologies like telemedicine, AI, and electronic health records (EHRs) is transforming healthcare delivery, enhancing patient management by providing real-time monitoring of chronic diseases, improving diagnostic accuracy, and enabling more effective treatment.

Goals & Challenges

Goals

Vision 2030 :

- Vision 2030 focuses on improving accessibility, quality , efficiency and sustainability
- Enhance healthcare services and foster preventative care measures
- Achieve universal health coverage by 2030 , ensuring comprehensive state-funded insurance for all citizens
- Increase private sector contributions from 40% to 65% in healthcare by 2030
- Foster healthcare innovation and research advancement to position Saudi Arabia as a leader in medical technology
- Enhance prevention and promote public health initiatives to reduce the burden of non communicable and chronic disease (1)

Challenges

- Growing burden in Non-communicable disease : High incidence of NCDs such as diabetes and obesity (2)
- Healthcare workforce shortage : Saudi needs an extra 175,000 healthcare workers by 2030 (3)
- Urban-rural healthcare disparities: Rural areas lack access to essential healthcare services (4)
- Challenges in primary healthcare :

1. Maldistribution of resources across different geographical locations
2. Shortage of health workforce
3. Lack of information systems

Reference:

1. Suleiman AK, Ming LC. Transforming healthcare: Saudi Arabia's vision 2030 healthcare model. Journal of Pharmaceutical Policy and Practice [Internet]. 2025 Jan 21 [cited 2025 Mar 23];18(1). Available from: https://pmc.ncbi.nlm.nih.gov/articles/PMC11753010/
2. Saeed A, Bin Saeed A, AlAhmri FA. Saudi Arabia Health Systems Challenging and Future Transformation With Artificial Intelligence. Cureus [Internet]. 2023 Apr 19 [cited 2025 Mar 23]; Available from: https://pmc.ncbi.nlm.nih.gov/articles/PMC10197987/
3. Global Health Exhibition. How Saudi's Vision 2030 is going to Transform the Healthcare Industry [Internet]. Globalhealthsaudi.com. 2025 [cited 2025 Mar 23]. Available from: https://www.globalhealthsaudi.com/en/news/healthcare-insights/how_saudis_vision_2030_is_going_to_transform_the_healthcare.html
4. emhj. The public health care system and primary care services in Saudi Arabia: a system in transition [Internet]. World Health Organization - Regional Office for the Eastern Mediterranean. 2020 [cited 2025 Mar 23]. Available from: https://www.emro.who.int/emhj-volume-26-2020/volume-26-issue-4/the-public-health-care-system-and-primary-care-services-in-saudi-arabia-a-system-in-transition.html

Financial Design

Saudi Arabia is the largest spender on healthcare across the Middle East and North Africa.
Health expenditure: 5.69% of GDP (2023).
Health expenditure per capita (US$): 1805, an increase from 906 in 2015.

Financing mix

- Mainly composed of public funding from the oil industry, with an additional contribution from the Value-Added Tax (VAT) introduced in 2018.
- The private sector contributes gradually through private insurance, OOP payments, and employer contributions.

Financing sustainability

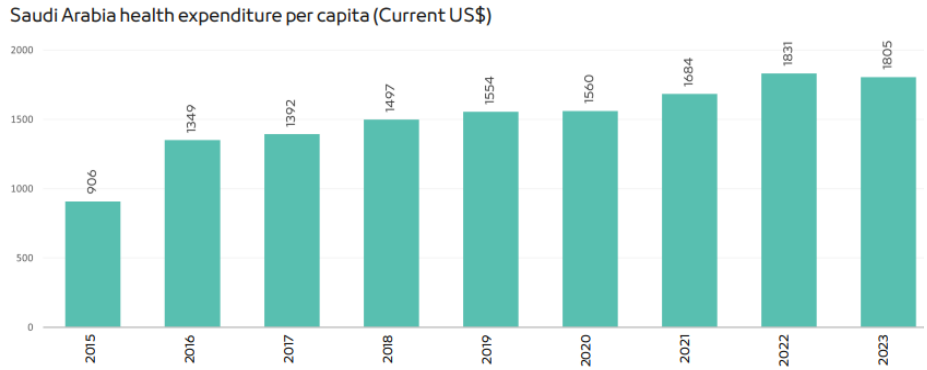
- Relatively sustainable due to a low debt-to-GDP ratio of 26.23%, indicating good fiscal health.
- Long-term sustainability is under threat from the rapid rise in health spending and global fluctuations in oil prices.

Extent of financial risk protection

The implementation of mandatory health insurance marked a significant step toward a more inclusive healthcare scheme. The policy ensures that a larger portion of the population will have access to quality healthcare without paying out-of-pocket expense, and it is a sign of commitment to improve healthcare equity and financial protection.

Item	2022	2023
Population	32,175,224	33,628,900
GDP (in SAR Millions)	4,157,143	4,003,436
GDP/Capita	129,203	119,047
THE (in SAR Millions)	220,910	227,684
THE/GDP	5.31%	5.69%
THE/Capita	6865	6770
GGHE (in SAR Millions)	160,841	156,918
GGHE/THE	72.81%	68.92%
GGHE/GDP	3.87%	3.92%
GGHE/Capita	4,999	4,666
OOPs (in SAR Millions)	28,240 ~	32,140 ~
OOPs/THE	12.78%	14.12%
Health Insurance (in SAR Millions)	31,829	38,625
Health Insurance / THE	14.41%	16.96%

Source: National Health Accounts 2022-2023



Source: National Health Accounts 2022-2023

Financial Design

Progressivity and fairness of the financing scheme

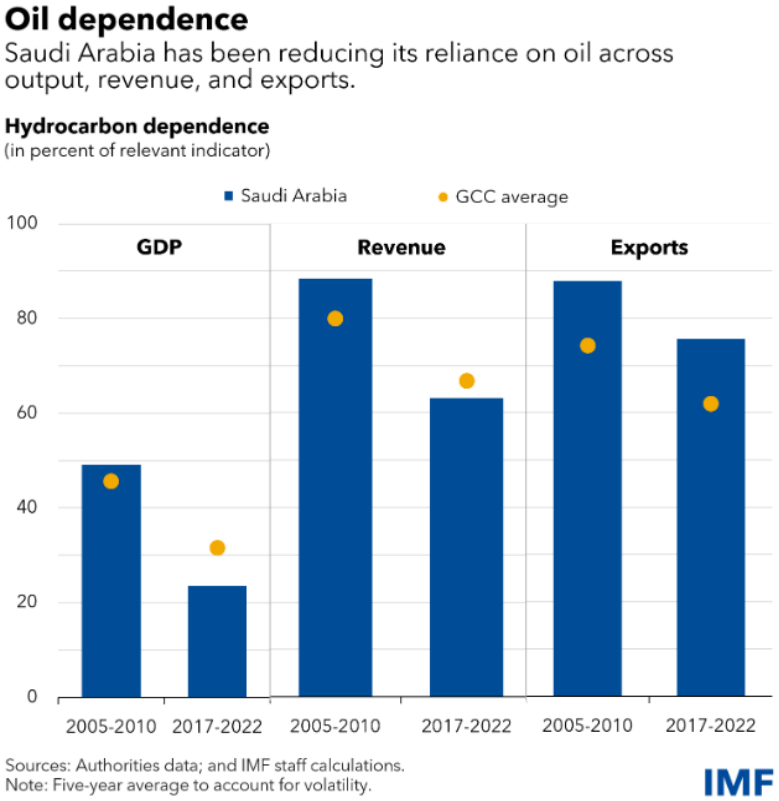
- Not highly progressive as the VAT is considered regressive. VAT tends to affect lower-income groups more than higher-income ones, raising societal concerns about fairness. The public funding from oil revenues may offset this to some extent but the overall fairness is questioned.

Pooling of funds

- Efficient pooling in the public sector, as the funds are pooled at the national level, ensuring healthcare access for everyone covered regardless of income or health risk.
- Less efficient/equal in the private sector, as the pooling is fragmented, with most insurance companies adopting risk-based pooling.

Vision 2030

- Vision 2030 aimed to transform the country's economic model by diversifying the income sources and reducing public spending with a greater focus on privatization and public-private partnerships.
- Significant progress was made with non-oil revenue increasing to 32% of total government revenue in 2022, up from less than 10% in 2012-13. The implementation of the VAT in 2018, along with the rate increase from 5% to 15% in 2020, has been the primary driver behind this growth in non-oil revenue.



Source: Authorities data; and IMF

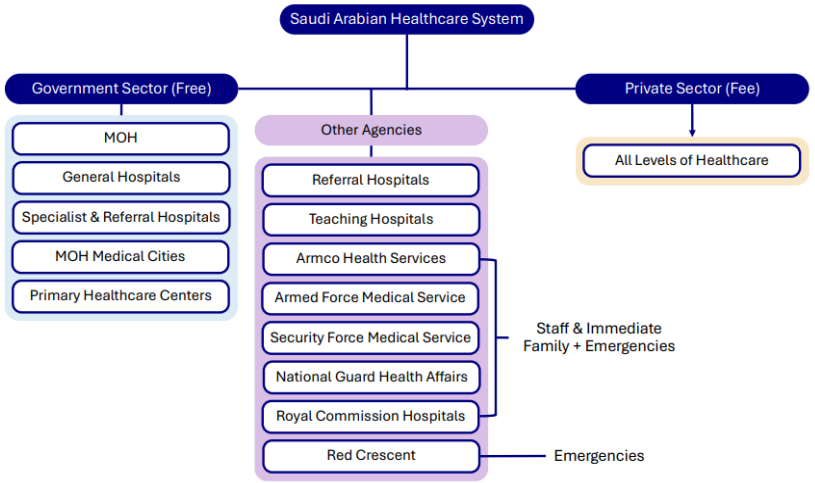
Stewardship & Organisation

Vertically integrated and segmented, mixed system
- The government controls everything from funding and policy-making to the provision of services and facilities.
- Four-tier (primary, secondary, tertiary, and quaternary) mixed system that includes both public and private sectors.

The Ministry of Health (MoH) is the primary service provider, financer, and regulator
- Balancing power: Powerful steward.
- Covers citizens, expatriates in the public sector, and pilgrims during Hajj and Umrah.

Other government entities provide healthcare through separate governance and budgets
- Facilities by the Armed Forces, National Guard, Ministry of the Interior, the Royal Commission of Jubail and Yanbu, university hospitals, and public enterprises like ARAMCO.
- Independently managed and funded through the budgets of their respective ministries.
- Compete for healthcare budget allocations with the MOH, leading to disparities in funding levels and policies.

Private healthcare insurance is compulsory for all employees in the private sector under the cooperative health insurance (CHI) programme.
- Both Saudi nationals and expatriates—as well as their dependents.
- Currently, the programme covers over 13 million citizens and expatriates.



Source: Ministry of Health, General Directorate of Hospitals

The MOH has a semi-decentralized structure
- 13 regions and 20 health directorates.
- Regional directorates have a degree of autonomy in planning and staffing; however, their limited financial authority hinders effective decision-making.

Vision 2030
- Moving towards a more decentralized structure, with the MOH shifting from its central role as the main payer, regulator, and service provider to focusing solely on regulation through privatization, corporatization, and digitalization ; and CHI will cover all citizens, residents, and visitors.

Service Delivery

Accessibility

- Variation of accessibility between urban and rural areas
- Out-of-pocket health expenditure decreased to 9.93% of total health spending in 2023
- Geographical isolation and limited service availability in rural areas (1)

Person-Centredness

- 81% of respondents were satisfied with outpatient responsiveness in 2024
- John Hopkins Aramco Helathcare revived Planetree Gold Certfifcate for excellence in Person-Centred care (2)

Coordination

- Decentralised system with 13 regions and 20 health Directories
- Ministry of Health (MOH) is the largest provider, accounting for over 62% of inpatient care
- Fragmentation in structure causes challenges for coordination of care pathways (3)

Accountably and efficiency

- Resource allocation varies between regions
- Healthcare spending increased from SAR 27 billion in 2005 to SAR 105 billion in 2023 showing commitment to improve healthcare system (4)

Quality

- Mandatory accreditation through Saudi Central Board for Accreditation of Health Institutions
- There is a focus on improving quality as part of vision 2030 goals (5)

Coverage

- Free healthcare for Saudi citizens through public system
- Plans for comprehensive state-funded insurance coverage for all citizens by 2026 (6)

Comprehensiveness

- Standard health insurance covers primary care, specialist services, hospital care and prescription medications
- Basic health insurance policy includes preventative, maternity care , and childcare (6)

References:

1. Abdullah A, Dhafer Abdullah Alshahrani, Majed Missfer Almalki, Alsharif RM, Abdulmageed Abdullatif Alhurani, Hamad Waslallah Aljrdahi, et al. Improving Access To Healthcare Services In Rural Areas Of Saudi Arabia. Migration Letters [Internet]. 2023 [cited 2025 Mar 23];20(S12):1540–5. Available from: https://migrationletters.com/index.php/ml/article/view/8832
2. eyeofriyadh. Johns Hopkins Aramco Healthcare (JHAH) Receives Highest Certification for Excellence in Person-Centered Care from Planetree [Internet]. eyeofriyadh.com. 2025 [cited 2025 Mar 23]. Available from: https://www.eyeofriyadh.com/news/details/johns-hopkins-aramco-healthcare-jhah-receives-highest-certification-for-excellence-in-person-centered-care-from-planetree
3. Kattan W. The state of primary healthcare centers in Saudi Arabia: A regional analysis for 2022. PLoS ONE [Internet]. 2024 Sep 9 [cited 2025 Mar 23];19(9):e0301918–8. Available from: https://pmc.ncbi.nlm.nih.gov/articles/PMC11383242/
4. Yazed AlRuthia, Abdulaziz N, Alorf NS, Yousif Asiri. How can Saudi Arabia reform its public hospital payment models? A narrative review. Saudi Pharmaceutical Journal [Internet]. 2020 Oct 6 [cited 2025 Mar 23];28(12):1520–5. Available from: https://pmc.ncbi.nlm.nih.gov/articles/PMC7537664/
5. Suleiman AK, Ming LC. Transforming healthcare: Saudi Arabia's vision 2030 healthcare model. Journal of Pharmaceutical Policy and Practice [Internet]. 2025 Jan 21 [cited 2025 Mar 23];18(1). Available from: https://pmc.ncbi.nlm.nih.gov/articles/PMC11753010/
6. Cigna. The health care system in Saudi Arabia [Internet]. Cigna Middle East. Cigna; 2023 [cited 2025 Mar 23]. Available from: https://www.cigna-me.com/en/blog/overview-healthcare-system-in-saudi-arabia

Provider Payments

Public Sector

- Global Budgeting - MOH healthcare facilities receive a fixed annual budget
- Salaried Payments - reduces financial incentives to overprovide services
- Service-Based Reimbursement - some specialist hospitals receive funding based on the volume and complexity of certain services

Private Sector

- Fee-for-service - reimbursed based on services rendered which can result in increased costs due to overutilization
- Insurance-Based Payments - many hospitals receive payments negotiated with insurers
- Capitation - emerging method some private hospitals are experimenting with

Vision 2030 Payment Reform

- Value-Based Payment - Saudi government shifting towards this model
- Diagnosis-Related Groups (DRGs) - Method being explored to standardize reimbursement and efficiency

Resource Allocation

In 2023, the Saudi government allocated approximately SR105 billion ($28.35 billion USD) to healthcare, accounting for 9.4% of its budget. However, a strategic goal of Vision 2030 is to reduce public healthcare expenditure by increasing private sector involvement.

- Privitisation
- Health Clusters
- Investment in Digital Health
- Focus on Non-communicable diseases
- Expanding number of hospital beds

Equity (in Access and Coverage)

- Saudi Citizens benefit from Universal Health Care
- Large disparity between urban and rural care, with more resources allocated to urban centres
- Expats not covered by UHC and can only access services if privately insured

Efficiency

- Increasing private sector contributions from 40% to 65% and adoption of new technologies is improving the efficiency of the healthcare system
- Decentralisation efforts into health clusters, is still in transition and not operating at full efficiency yet.

Effectiveness

- Heavy reliance on expat healthcare workers impacts service continuity and training programs.
- High doctor-patient ratios affect service consistency across regions.

Responsiveness

- The healthcare system is subject to long wait times in public hospitals but with the increasing privitisation it will reduce the burden on the public healthcare system.

Recommendations

Improve Saudization of Local Healthcare Workforce

- Expand nursing and medicine education through regional and corporate partnerships with the Ministry of Education, utilising local development funds.
- Introduce family-friendly policies, flexible shifts, and childcare support, while deploying allied health professionals to reduce nursing workloads.
- Diversifying incentive strategies (access to public land, increasing salary, bonus for night shifts e.g.,).

Improve Resource Allocation

Improve resource allocation in rural areas :

- The decentralisation of the healthcare system has created greater health inequalities in rural areas.
- They had reduced access to specialist care and fewer hospital beds.
- By increasing telehealth and communications with rural healthcare centres it can help reduce the inequality in care.

Improving Expatriate Workers' Conditions and Enhancing Recruitment Standards

- Standardise recruitment processes with international agencies to hire highly skilled workers, avoiding tokenism.
- Provide cultural and language training to support expatriate workers' integration.
- Implement corporate policies promoting a positive working environment and discourage violence, particularly towards expatriate workers.

Mental Health School and Social Awareness Education

- Launch social awareness campaigns via social media and podcasts to combat mental health stigma and encourage seeking treatment.
- Integrate mental health education into school curricula, emphasizing it as a treatable condition, not a stigma, and deploy school-based mental health nurses for early support.

Thank You For Listening